PRECALCULUS · UNIT 6: MATRICES

Spy Agency Codes

Teacher Guide & Answer Keys

Teacher

2–3 class periods

Project Overview

Students work for a spy agency. Gadget inventories use matrix addition, scalar multiplication, and products; secret messages are encoded with a key matrix and decoded with its inverse; and supply problems are systems solved with inverse matrices.

The determinant decides whether a code can be broken.

Standards Addressed

- N-VM.C.6–8 Matrices to represent data; add, subtract, scalar multiply, multiply
- N-VM.C.10 Determinants and inverses
- A-REI.C.8–9 Matrix equations and inverse matrices
- Applications to codes

Pacing (2–3 class periods)

Day	Plan
Day 1	Launch with a coded message on the board. Part 1 inventory matrices.
Day 2	Part 2 encoding and decoding.
Day 3	Part 3 systems and the mission file.

Materials

- Student handout and Inventory Cards
- Graphing calculators with matrix menu
- Envelopes for coded messages

How the Three Versions Differ

Level	What changes
CP2	Adding and scaling matrices, matrix-times-column products, and encoding/decoding one word.
CP1	Values after changes, determinants and inverses, multi-pair messages, and a 2 × 2 system.
Honors	Transfer matrices, cofactor determinants, designing keys, decoding intercepts, and a 3 × 3 system.

Watch For (Common Misconceptions)

- Multiplying matrices entry by entry.
- Ignoring dimension rules.
- Multiplying by the inverse on the wrong side.

Extension Ideas

- Exchange coded messages with another group.
- Research the Hill cipher.
- Use mod 27 arithmetic to keep codes in the alphabet.

Answer Key – CP2

Part 1: Inventory

#	Task	Answer
1	Write matrix I and give its dimensions.	[[12, 5, 3], [8, 9, 4], [15, 6, 7]]; 3 × 3
2	Add the New Shipment matrix S = [[2, 1, 1], [2, 1, 1], [2, 1, 1]] to I.	[[14, 6, 4], [10, 10, 5], [17, 7, 8]]
3	The agency doubles the original inventory. Find 2I.	[[24, 10, 6], [16, 18, 8], [30, 12, 14]]
4	What does the entry in row 3, column 3 of I mean?	Tokyo has 7 drones.

Part 2: Costs

#	Task	Answer
5	Find the value of the Boston gadgets: 12(40) + 5(250) + 3(900).	480 + 1,250 + 2,700 = $4,430
6	Find I · C (the value of each warehouse).	Boston $4,430, London $6,170, Tokyo $8,400
7	Which warehouse has the most valuable inventory?	Tokyo
8	Why can we multiply a 3 × 3 matrix by a 3 × 1 matrix?	The inner dimensions (3 and 3) match; the result is 3 × 1.

Part 3: Secret Codes

#	Task	Answer
9	Write the word HI as a column matrix.	[8, 9]
10	Encode HI by computing K · [8, 9].	[3(8) + 9, 5(8) + 2(9)] = [33, 58]
11	Decode [33, 58] using K^{-1} = [[2, −1], [−5, 3]]. Do you get HI back?	[2(33) − 58, −5(33) + 3(58)] = [8, 9] → HI ✓

Reflection

#	Prompt	Look for
R1	Why do the dimensions have to match to add matrices?	Each entry is added to the matching entry, so the shapes must be the same.
R2	Why does the inverse matrix decode the message?	$K^{-1}K$ = I, so multiplying by K^{-1} undoes the encoding.

Answer Key – CP1

Part 1: Inventory & Value

#	Task	Answer
1	Find I + S and the value of each warehouse after the shipment.	I + S = [[14, 6, 4], [10, 10, 5], [17, 7, 8]]; values $5,660, $7,400, $9,630
2	Prices rise 10%. Write the new cost matrix and find the new Tokyo value for the original inventory.	1.1C = [44, 275, 990]; Tokyo 15(44) + 6(275) + 7(990) = $9,240
3	Is I · C the same as C · I? Explain using dimensions.	No. I (3 × 3) times C (3 × 1) gives a 3 × 1 matrix of warehouse values; C (3 × 1) times I (3 × 3) is undefined because the inner dimensions (1 and 3) do not match.
4	Find the total of each gadget across all warehouses using [1, 1, 1] · I.	[35, 20, 14]

Part 2: Encode & Decode

#	Task	Answer
5	Find det K and K^{-1}.	det = 6 − 5 = 1; K^{-1} = [[2, −1], [−5, 3]]
6	Encode the message GO NOW in pairs (GO, _N, OW).	GO [7, 15] → [36, 65]; _N [27, 14] → [95, 163]; OW [15, 23] → [68, 121]
7	Decode the message [58, 97], [23, 40].	[2(58) − 97, −5(58) + 3(97)] = [19, 1] → S, A; [2(23) − 40, −5(23) + 3(40)] = [6, 5] → F, E. The message is SAFE.
8	Why must the key have a nonzero determinant?	Otherwise it has no inverse and messages cannot be decoded.

Part 3: Supply Systems

#	Task	Answer
9	A shipment of pens and watches has 30 items and costs $3,510. Write a matrix equation and solve it with an inverse matrix.	[[1, 1], [40, 250]] · [p, w] = [30, 3,510]; det = 210; [p, w] = $\frac{1}{210}$[[250, −1], [−40, 1]] · [30, 3,510] = [19, 11]
10	Check your answer by substitution.	19 + 11 = 30 ✓; 19(40) + 11(250) = 760 + 2,750 = 3,510 ✓
11	Write your mission file with inventory, a coded message for a partner, and the system you solved.	Answers vary.

Reflection

#	Prompt	Look for
R1	What does the determinant tell you about a matrix?	If it is 0 the matrix has no inverse; otherwise an inverse exists.

#	Prompt	Look for
R2	How is solving AX = B with an inverse like solving ax = b?	Multiply both sides by the inverse (like dividing by a) to isolate X.

Answer Key – Honors

Part 1: Inventory Analysis

#	Task	Answer
1	A transfer matrix T moves 20% of Boston's gadgets to London each month. Write the new inventory after one month as a matrix product or sum.	Boston row × 0.8, London row + 0.2 × Boston row: [[9.6, 4, 2.4], [10.4, 10, 4.6], [15, 6, 7]]
2	Find the total value matrix after the transfer. Did the total value change?	Values: Boston 3,544, London 7,056, Tokyo 8,400; total unchanged at $19,000
3	Explain why matrix multiplication is not commutative using this context.	Order matters: (warehouses × gadgets)(gadgets × costs) gives values; the reverse product is undefined or means something else.
4	Find det I using cofactor expansion. Is I invertible?	12(63 − 24) − 5(56 − 60) + 3(48 − 135) = 468 + 20 − 261 = 227; yes

Part 2: Code Design

#	Task	Answer
5	Design a 2 × 2 key matrix with determinant 1 (so the inverse has integer entries). Give K and K^{-1}.	Answers vary; e.g., K = [[2, 3], [1, 2]], K^{-1} = [[2, −3], [−1, 2]]
6	Encode MEET with your key and decode it to check.	Answers vary by key; with the example: ME [13, 5] → [41, 23]; ET [5, 20] → [70, 45]
7	An intercepted message was encoded with K = [[3, 1], [5, 2]]: [39, 70], [44, 75]. Decode it.	[2(39) − 70, −5(39) + 3(70)] = [8, 15] → H, O; [2(44) − 75, −5(44) + 3(75)] = [13, 5] → M, E. The message is HOME.
8	Explain why a key with determinant 2 would cause trouble with whole-number codes.	Its inverse contains fractions (½), so decoding may not give whole numbers easily.

Part 3: 3 × 3 Supply System

#	Task	Answer
9	Write the system and the matrix equation.	p + w + d = 27; 40p + 250w + 900d = 4,920; p − 2w − 2d = 0 → [[1, 1, 1], [40, 250, 900], [1, −2, −2]] · [p, w, d] = [27, 4,920, 0]
10	Solve using an inverse matrix (technology allowed) and check.	p = 18 pens, w = 6 watches, d = 3 drones (18 + 6 + 3 = 27; 720 + 1,500 + 2,700 = 4,920; 18 = 2(9))
11	Write the mission file with inventory analysis, your code system, and the supply solution.	Answers vary.

Reflection

#	Prompt	Look for
R1	Why does an invertible key matter for a code?	Only an invertible matrix can be undone to recover the message.
R2	How did you check a matrix solution?	Substitute back into every original equation (or multiply A by X).